DynoX thrust stand

RTS Pro 20

User manual · 20 kgf thrust and torque test stand



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Designed and manufactured by REUDE Technologies Private Limited, Chennai, India. Read the safety instructions in Section 3 before assembling or operating this equipment.

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1. About this manual

This manual explains how to assemble, wire, calibrate, operate and maintain the RTS Pro 20. It is written for technicians, engineers and lab staff with basic mechanical and electrical skills. Keep it with the stand and pass it on with the equipment.

How to read the safety notes

WARNING A hazard that can cause serious injury. Follow the instruction exactly.

CAUTION A risk of damage to the stand, the sensors or the measurement.

NOTE Useful information that makes the job easier or the result better.

Interactive 3D model

Every component in this manual is numbered the same way as in the DynoX 3D viewer. Open the viewer to rotate the stand, see the exploded sub-assemblies, and lay every component out on the table.

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2. Product overview

The RTS Pro 20 is a floor-standing thrust stand for heavy-lift drone propulsion up to 20 kgf of thrust. Twin 1,000 mm pillars on a 1,000 mm base, braced at the front, rear and both sides, hold the motor about 1.08 m above the floor, clear of ground effect. A three-cell sensing head (one upright thrust cell and two torque cells) measures thrust and torque together, while the ROTRIX DAQ unit records RPM, voltage, current, vibration and temperature in sync and streams them to the ROTRIX software.

Specifications

Item	Specification
Rated thrust	0 to 20 kgf
Measured parameters	Thrust, torque, RPM, voltage, current, vibration and temperature
Force sensing	Three beam load cells: one upright thrust cell and two horizontal torque cells; ±0.5% of full scale
Motor interface	Stainless-steel motor plate, 200 × 207 mm, across the two torque cells
Speed sensing	Optical sensor (QRD1114) on a printed arm, 0 to 25,000 RPM, ±1%
Electrical sensing	Inline current sensor module on the right pillar (5 to 100 A); voltage 3S to 12S measured by the DAQ
Vibration and temperature	Accelerometer board behind the motor plate, ±16 g; temperature -55 to 125 °C
Data acquisition	ROTRIX DAQ unit on the middle pillar bridge; STM32 controller, signal conditioning and protection; USB link
Software	ROTRIX test control and logging; ROTRIDASH analytics (optional)
Frame	40-series aluminium extrusions with orange slot covers (REUDE standard)
Footprint	About 845 × 1,000 mm including the side outriggers
Thrust axis height	About 1,080 mm above the floor
Reference propeller	40 inch (about 1,030 mm) two-blade carbon fibre

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Main components

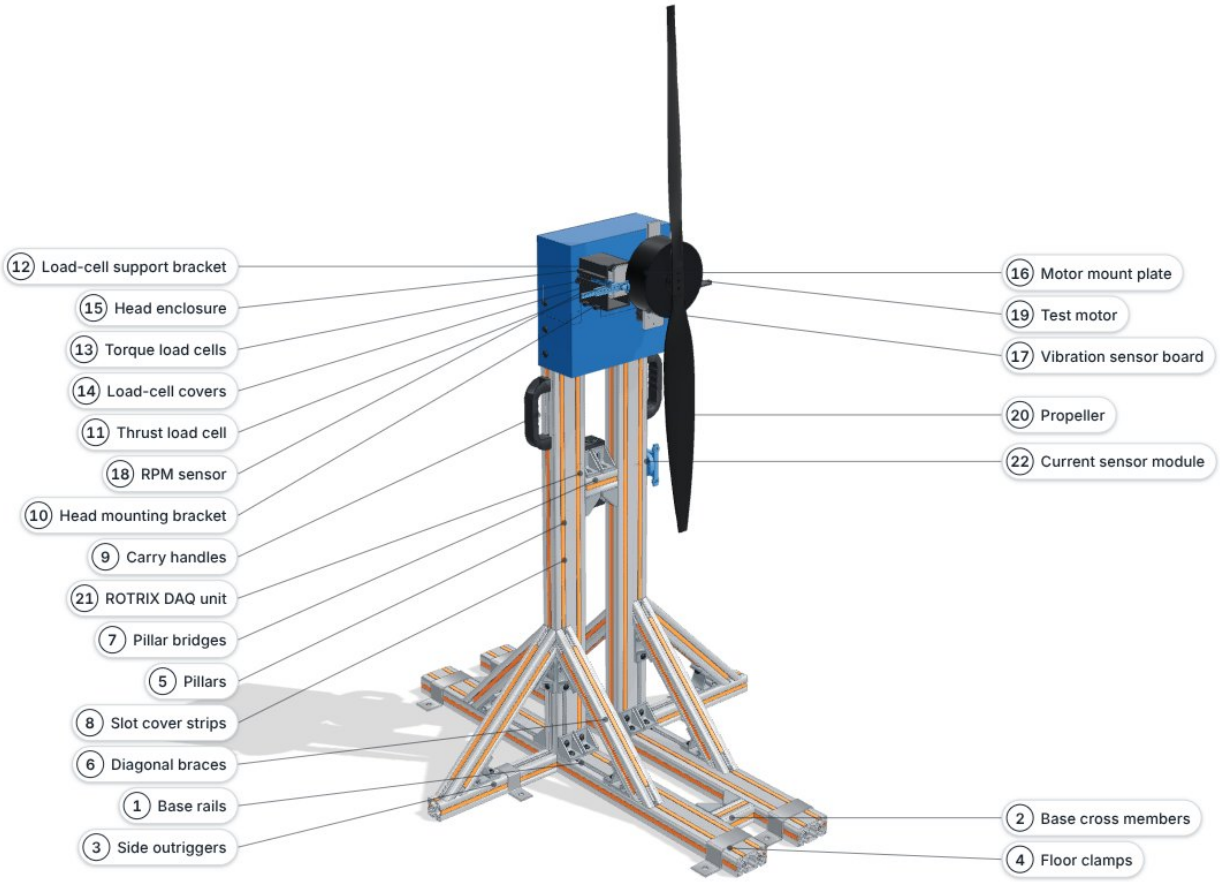


Figure 2.1 RTS Pro 20 assembled, with the main components numbered.

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Exploded view

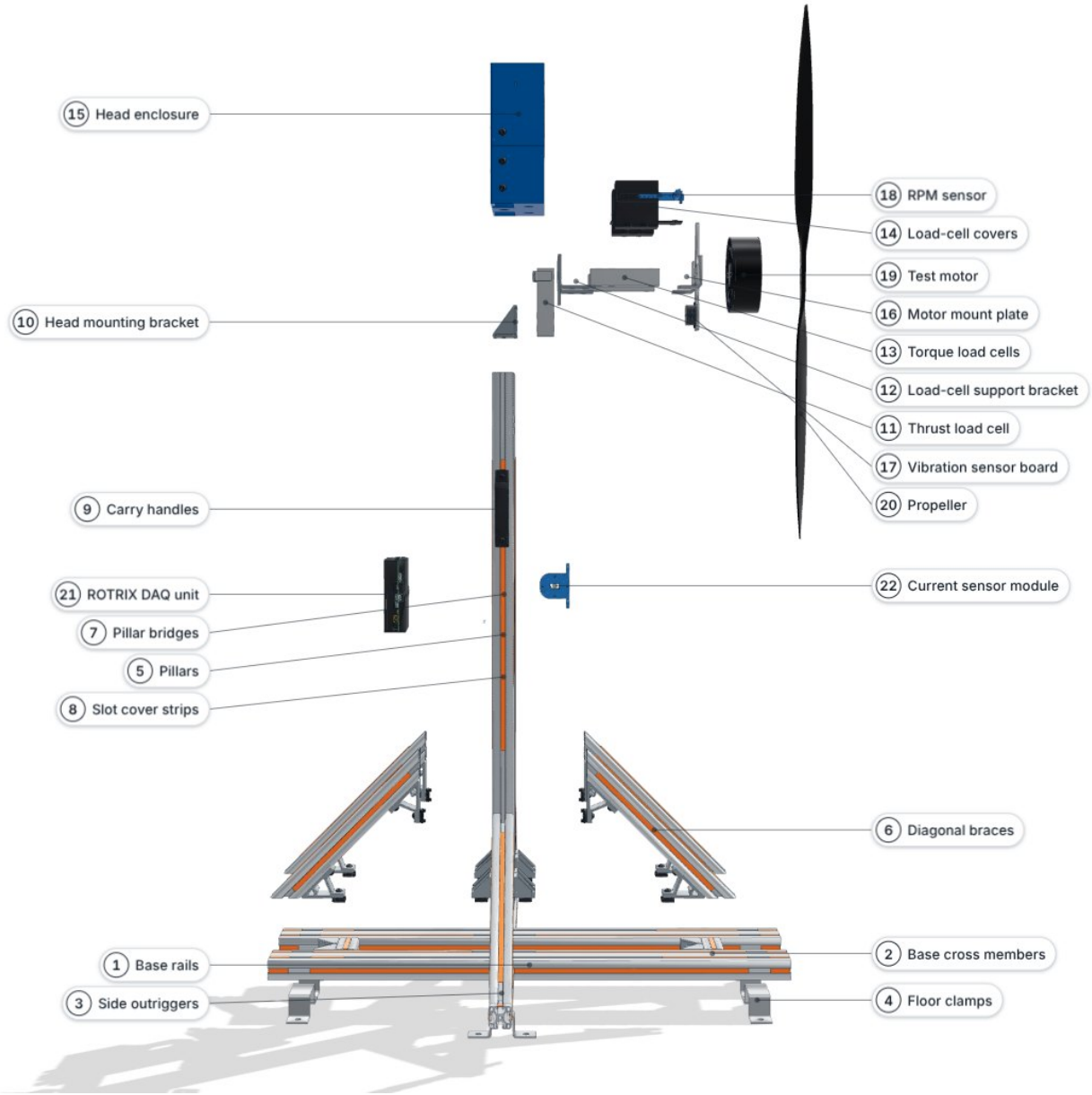


Figure 2.2 Exploded side view. Each sub-assembly moves as one block, in the order it is assembled.

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3. Safety instructions

WARNING A spinning propeller can cause severe injury. Never approach the stand while a battery is connected, and keep everyone out of the propeller plane during a test.

Assembly team

At least two people are required, and three when standing the pillars.

People and protection

- Read this manual completely before assembling or operating the stand.
- Wear safety glasses and gloves during assembly. Wear safety glasses and hearing protection during every test run.
- Tie back long hair and remove loose clothing, lanyards and jewellery before approaching a running motor.
- Never reach over, under or past a propeller while a battery is connected, even if the motor is stopped.

Test area

- Place the stand on a level, rigid floor or bench and anchor it with the supplied clamps before any powered test.
- Mark an exclusion zone around the stand. Nobody may stand in the plane of the propeller disc while a battery is connected.
- Keep the area clear of loose paper, tools, cable offcuts and anything that the propeller wash can lift.
- Test indoors or under cover. The sensors and the DAQ unit are not weatherproof: keep them away from rain, condensation and dust.

Electrical and battery

- Connect the battery last and disconnect it first. Keep the throttle at zero and the ESC disarmed while connecting.
- Use cables, connectors and fuses rated above the maximum test current. Check insulation before every session.
- Fit an emergency-stop or kill switch in the power line, within reach of the operator, and test it before each session.
- Charge, store and transport lithium batteries in fire-safe bags or containers, and never leave a charging battery unattended.
- When the operator works at a distance, only the digital data link and the emergency-stop loop should run to the operator station. Do not extend the analogue sensor cables.

Operation

- Never exceed the rated thrust of the stand. Stop the test if a reading approaches the limit.
- Before each run, check that every bolt is tight, the propeller is undamaged and balanced, and its hub bolts are secure.
- Start every new motor and propeller combination at low throttle and increase in steps while watching vibration and temperature.
- Stop immediately on unusual noise, rising vibration, smoke, or any reading that does not make sense.

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- Do not lean on, lift or carry the stand by the sensing head. Loads on the head can permanently damage the load cells.

CAUTION The rated thrust of this stand is 20 kgf. Loads above the rating can permanently damage the load cells and void the calibration.

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4. Package contents

Check every item against this list before you start. If anything is missing or damaged, contact REUDE support before assembly. Do not start with incomplete or faulty parts.

No.	Component	Code	Size	Material	Qty
1	**Base rails**	40 × 80 aluminium extrusion	40 × 80 mm section, 1,000 mm long	Anodised 6063 aluminium extrusion	2
2	**Base cross members**	40 × 40 profile	40 × 40 × 85 mm, two 90° brackets each	Anodised aluminium extrusion	2
3	**Side outriggers**	40 × 40 profile	40 × 40 × 300 mm	Anodised aluminium extrusion	2
4	**Floor clamps**	U-clamps and clamp bars	U-clamps 165 × 40 × 43 mm, clamp bars 125 × 30 × 43 mm	Zinc-plated steel	4 U-clamps, 2 bars
5	**Pillars**	40 × 80 aluminium extrusion	40 × 80 × 1,000 mm	Anodised 6063 aluminium extrusion	2
6	**Diagonal braces**	40 × 40 profile at 45°	40 × 40 mm profile, about 400 mm long, 45° brackets at both ends	Anodised aluminium extrusion	6
7	**Pillar bridges**	40 × 40 profile	40 × 40 × 85 mm	Anodised aluminium extrusion	2
8	**Slot cover strips**	Orange T-slot covers · REUDE standard	9 mm wide, cut to length for each slot	Flexible PVC, REUDE orange	1 set
9	**Carry handles**	Polyamide grips	145 mm long	Glass-filled polyamide, M8 screws	2
10	**Head mounting bracket**	5 mm vertical support	80 × 45 × 60 mm, 5 mm plate with two triangular gussets	Laser-cut steel, powder coated	1
11	**Thrust load cell**	Vertical beam cell with clamp	32 × 32 × 130 mm, sheet-metal clamp on top	Alloy-steel strain-gauge load cell, steel clamp	1
12	**Load-cell support bracket**	5 mm sheet-metal cradle	200 × 90 × 70 mm	Laser-cut and folded steel	1
13	**Torque load cells**	Horizontal beam cell pair	32 × 32 × 130 mm each, 164 mm apart	Alloy-steel strain-gauge load cells	2
14	**Load-cell covers**	Side guards	76 × 86 × 80 mm each	3D-printed PETG	2
15	**Head enclosure**	Printed front and laser-cut base	251 × 260 × 100 mm	3D-printed panels with a laser-cut base plate	1 set of 3
16	**Motor mount plate**	Motor interface	200 × 207 mm	Stainless steel	1
17	**Vibration sensor board**	ADXL345 accelerometer	51 × 50 × 18 mm	FR-4 board in a printed holder	1
18	**RPM sensor**	QRD1114 optical sensor	105 mm arm with sensor head	3D-printed arm and housing	1

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No.	Component	Code	Size	Material	Qty
19	**Test motor**	RNG QB20 (reference)	Ø145 × 56 mm	Brushless DC motor	1
20	**Propeller**	40 inch carbon fibre	1,026 mm diameter	Carbon-fibre composite	1
21	**ROTRIX DAQ unit**	STM32-based acquisition	125 × 140 × 41 mm	Custom PCB in a printed enclosure	1
22	**Current sensor module**	Inline current sensor	24 × 90 × 58 mm	3D-printed case, current sensor board	1

Brackets and fixings

No.	Fixing	Code	Size	Material	Qty
23	**90° corner brackets**	40 series	40 × 40 × 35 mm	Die-cast aluminium	16
24	**45° brace brackets**	Diagonal end brackets	35 × 58 × 80 mm	Die-cast aluminium	12
25	**T-slot nuts**	Series 8	20 × 25 mm	Printed PLA tee nuts and steel slot nuts	about 50
26	**M8 screws**	Button and socket head	M8 × 16 mm	Steel, black oxide	about 50

Component importance

Critical parts are in the load path or the measurement chain. **Important** parts set the stiffness and alignment of the frame. **Standard** parts are for convenience and protection.

No.	Component	Importance	Why it matters
1	Base rails	**Critical**	They set the footprint along the thrust line. Twenty kilograms of thrust acts about one metre above the floor, and the rails are what stop it tipping the stand forwards.
2	Base cross members	**Important**	They keep the rails parallel and square so the tower stands true and the thrust line stays straight.
3	Side outriggers	**Important**	Motor torque tries to roll the stand sideways. The outriggers give it a wide footprint so it stays planted at full power.
4	Floor clamps	**Critical**	Thrust tries to lift and slide the stand. Anchoring it keeps the readings clean and the test area safe.
5	Pillars	**Critical**	They hold the propeller axis well above the floor, away from ground effect, and carry all the thrust down into the base. Pillars out of square tilt the thrust line.
6	Diagonal braces	**Critical**	Triangles are the stiffest shape in a frame. The braces stop sway and damp vibration so the load cells read thrust, not wobble.
7	Pillar bridges	**Important**	They stop the pillars spreading or twisting under load. The top bridge is the seat for the sensing head.
8	Slot cover strips	**Standard**	They keep dust and small parts out of the slots, cover sharp edges, hold cables in place and give the stand its orange accent.
9	Carry handles	**Standard**	They make the stand safer to move. Handles sit outside the load path and do not affect measurement.

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No.	Component	Importance	Why it matters
10	Head mounting bracket	Critical	It is the only link between the sensors and the frame. Any play here shows up as noise in thrust and torque.
11	Thrust load cell	Critical	This is the cell that measures thrust. It is factory calibrated, and every thrust figure the stand produces comes from it.
12	Load-cell support bracket	Critical	It carries thrust into the thrust cell and splits motor torque between the two torque cells.
13	Torque load cells	Critical	Torque, with speed, gives shaft power and efficiency. The pair cancels any thrust that leaks across.
14	Load-cell covers	Standard	They keep fingers, cables and propeller wash off the cells.
15	Head enclosure	Standard	It shields the thrust cell and its wiring from dust, knocks and propeller wash.
16	Motor mount plate	Critical	A stiff plate keeps the motor square to the thrust axis, so all of the force reaches the cells.
17	Vibration sensor board	Important	It catches imbalance and loose parts early, before they damage the motor or the stand.
18	RPM sensor	Important	Speed is needed for efficiency figures and thrust-versus-RPM curves.
19	Test motor	Important	This is the unit under test. Any motor within the 20 kgf rating fits the plate.
20	Propeller	Critical	The tips move very fast at full power. Good balance, tight hub bolts and a clear zone are essential for safety and clean data.
21	ROTRIX DAQ unit	Critical	It samples thrust, torque, speed, voltage, current, vibration and temperature together and sends them to your PC over USB.
22	Current sensor module	Important	Current, with voltage, gives electrical power, so it is needed for every efficiency figure.

All components laid out

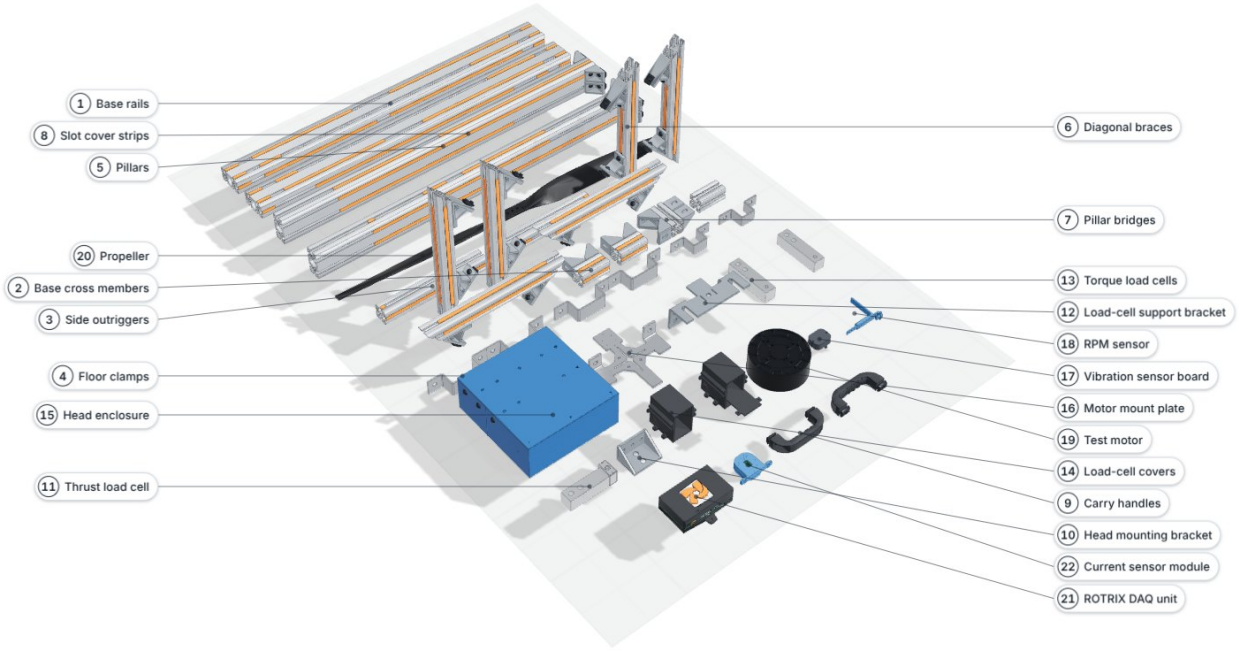


Figure 4.1 The components laid out on the table, as shipped.

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5. Tools and tightening torques

Tools you will need

- Hex keys 4, 5 and 6 mm
- Spanner or socket set
- Torque wrench
- Spirit level, try square and tape measure
- Cable ties, a side cutter and a sharp knife (for the slot covers)
- Multimeter and clamp meter
- Hand tachometer (for the RPM check)
- Calibration weights (for example 2, 5 and 10 kg) with cord and a low-friction pulley
- Masonry drill and M10 anchors for concrete floors

Recommended tightening torques

Fastener	Where used	Typical tightening
M3 screws	Printed parts, sensor boards, DAQ enclosure	Hand tight, about 0.5 to 1 Nm. Do not crack the printed parts.
M6 bolts	Motor to mounting plate, plate to bracket	About 8 Nm, with thread-locker on motor bolts
M8 button-head bolts	Brackets into T-slot nuts	About 12 to 15 Nm into steel T-nuts; snug only (about 4 Nm) into printed T-nuts. Tighten evenly around each joint.

NOTE Tighten in two passes: first snug every bolt in a joint, then bring each to the final torque. This keeps the frame square.

CAUTION Always load the T-slot nuts into a profile before a bracket, end cap or another profile closes that slot. A missing nut means taking the joint apart again.

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6. Assembly, step by step

Work through the steps in order. In each figure the parts added in that step are highlighted in orange; parts fitted in earlier steps are shown in their final position.

Step 1. Base rails and cross members



Figure 6.1 Base rails and cross members.

1. Lay the two 1,000 mm base rails (40 × 80 mm) parallel on a level floor, 85 mm apart.
2. Load T-nuts into the rail top slots for the pillar feet, the six brace feet, the outriggers and the cross members.
3. Fit a 90° bracket to each end of the two 85 mm cross members.
4. Place one cross member between the rails at the front and one at the rear, flush with the rail ends, and bolt them.
5. Measure both diagonals of the frame. They should match within 2 mm. Adjust, then tighten.

CAUTION Load every T-nut the rails need now. They cannot be added once brackets close the slots.

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Step 2. Pillars

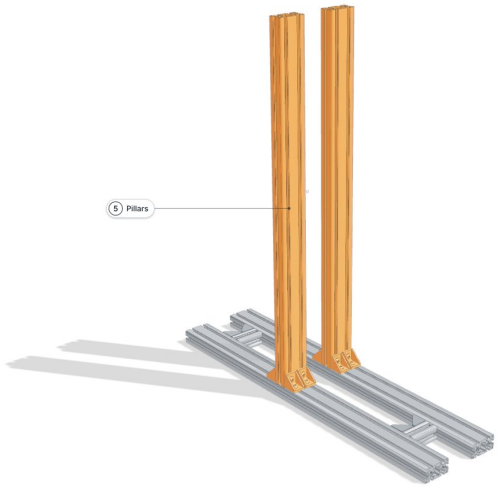


Figure 6.2 Pillars.

1. Load T-nuts into both pillars for the bridges, the brace tops, the handles, the head enclosure and the current sensor.
2. With two people holding, stand each 1,000 mm pillar on its rail, 430 mm from the rear end, wide face towards the propeller.
3. Fix each pillar foot with four 90° corner brackets and M8 screws.
4. Check that both pillars are vertical on two faces and parallel to each other, then tighten.

CAUTION A pillar can topple before its brackets are tight. Keep two people holding the pillars until all eight brackets are bolted.

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Step 3. Side outriggers and floor clamps

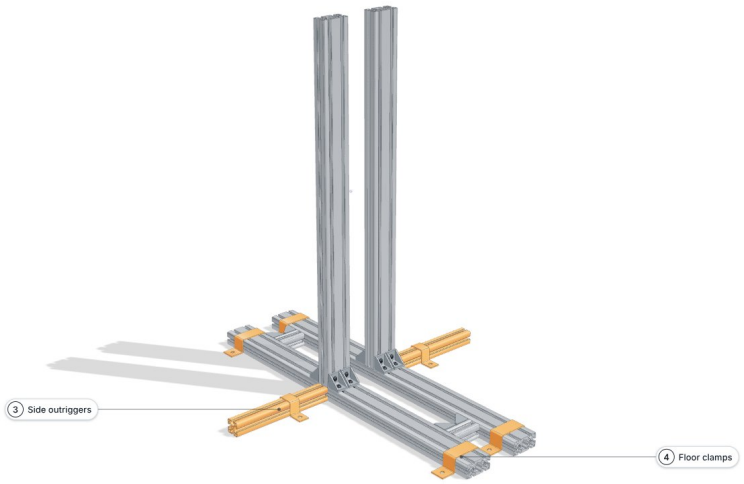


Figure 6.3 Side outriggers and floor clamps.

1. Butt a 300 mm outrigger against the outer face of each rail, in line with the pillars, and fix it with a 45° bracket.
2. Move the frame to its final position.
3. Place a U-clamp over each rail end and a clamp bar over each outrigger end. Mark the anchor holes, but leave the clamps loose until Step 5.

Step 4. Diagonal braces

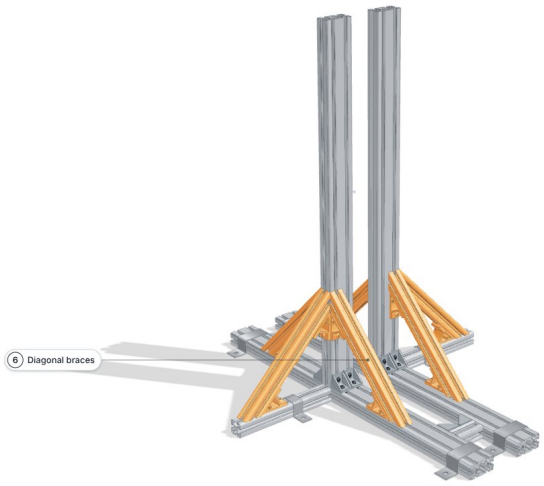


Figure 6.4 Diagonal braces.

1. Fit a 45° bracket at each end of all six braces.
2. Bolt the two front braces and the two rear braces between the rails and the pillars.
3. Bolt the two side braces between the outriggers and the pillars.
4. Tighten all brace screws evenly, then recheck that the pillars are vertical.

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Step 5. Pillar bridges, handles and anchoring

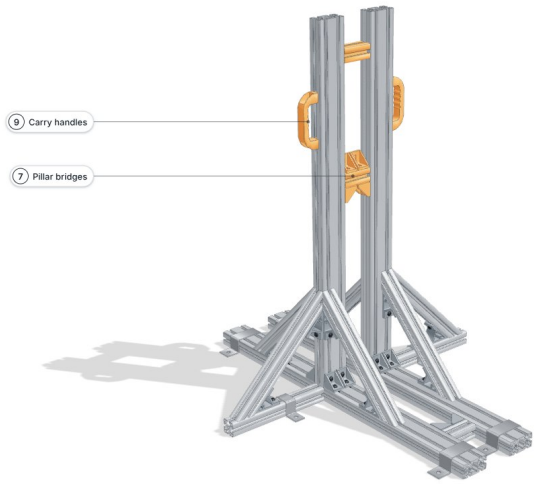


Figure 6.5 Pillar bridges, handles and anchoring.

1. Fit the top bridge flush with the pillar tops and the middle bridge about 600 mm above the floor, each with 90° brackets.
2. Screw a carry handle to the outer slot of each pillar, about 700 mm above the floor.
3. Tighten the whole frame to the recommended torque.
4. Drill, anchor and tighten all six floor clamps.

CAUTION Anchor the clamps only after the frame is fully squared and tightened. The stand must be anchored before any powered test.

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Step 6. Head mounting bracket and thrust load cell



Figure 6.6 Head mounting bracket and thrust load cell.

1. Bolt the 5 mm head mounting bracket to the top bridge through two T-nuts and check it is square to the pillars.
2. Bolt the thrust load cell upright to the front face of the bracket, cable end down.
3. Fit the sheet-metal clamp over the top end of the load cell.

CAUTION Handle the load cells gently. Never use the head as a handle and never load it by hand beyond the rated thrust.

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Step 7. Load-cell support bracket and torque load cells

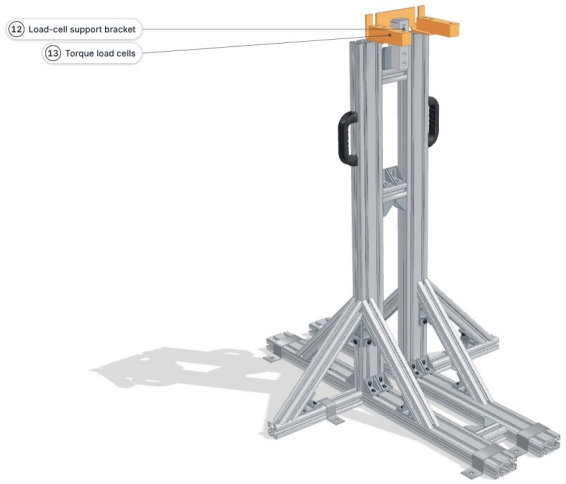


Figure 6.7 Load-cell support bracket and torque load cells.

1. Bolt the folded load-cell support bracket to the clamp on top of the thrust cell.
2. Bolt one torque load cell to each side of the support bracket, pointing forward.
3. Check that both cells are level and parallel to the thrust axis.

Step 8. Motor plate, vibration board, motor, RPM sensor and propeller

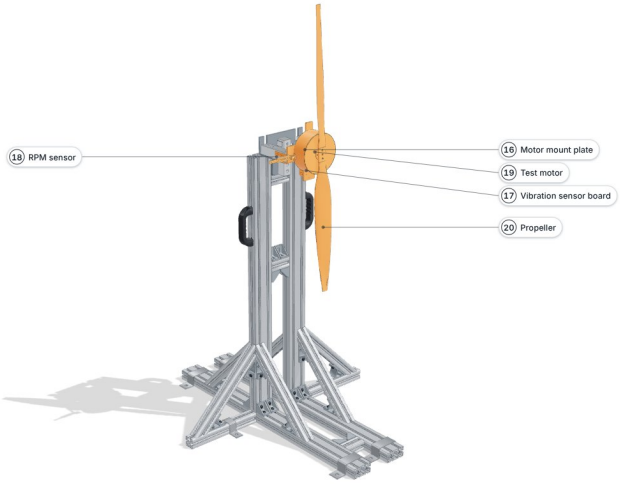


Figure 6.8 Motor plate, vibration board, motor, RPM sensor and propeller.

1. Bolt the stainless motor plate across the front ends of the two torque cells.
2. Screw the vibration board holder to the back of the motor plate.
3. Bolt the motor to the plate with thread-locker and check that the shaft is centred.
4. Fix the RPM sensor arm on the left load-cell cover and aim the sensor at the blade path.
5. Fit the propeller last, after wiring is complete and with the battery disconnected.

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Step 9. DAQ unit, current sensor, covers and head enclosure

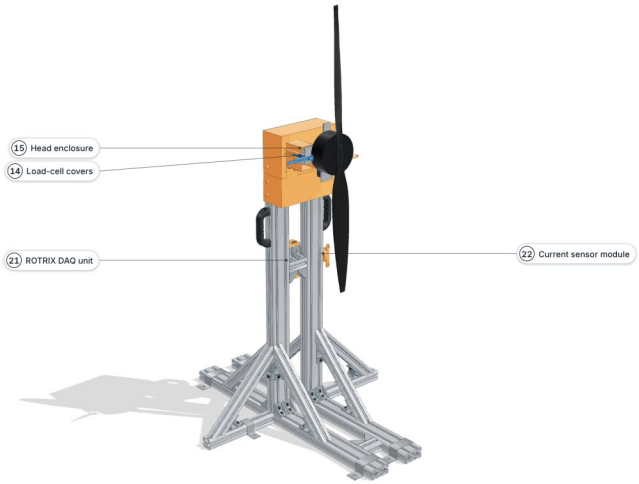


Figure 6.9 DAQ unit, current sensor, covers and head enclosure.

1. Screw the ROTRIX DAQ unit to the rear of the middle pillar bridge, connectors facing up.
2. Screw the current sensor case to the outer slot of the right pillar, below the handle.
3. Complete the wiring in Section 8.
4. Clip the two load-cell covers over the torque cells, then fit the head enclosure and screw it to both pillars without trapping any cable.

CAUTION The DAQ unit is supplied assembled and sealed. Do not open the enclosure.

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Step 10. Orange slot cover strips (REUDE standard)



Figure 6.10 Orange slot cover strips (REUDE standard).

1. Check that every bolt in the frame is at its final torque and that all cables are routed.
2. Press an orange cover strip into each open T-slot, starting at one end and working along by hand.
3. Cut the strip flush at each bracket and at the profile ends with a sharp knife.
4. Where a cable runs inside a slot, lay it in the slot first and press the cover over it.

CAUTION Fit the covers last. They are easy to lift out later if you need to add a T-nut or re-route a cable.

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Sensing head, exploded

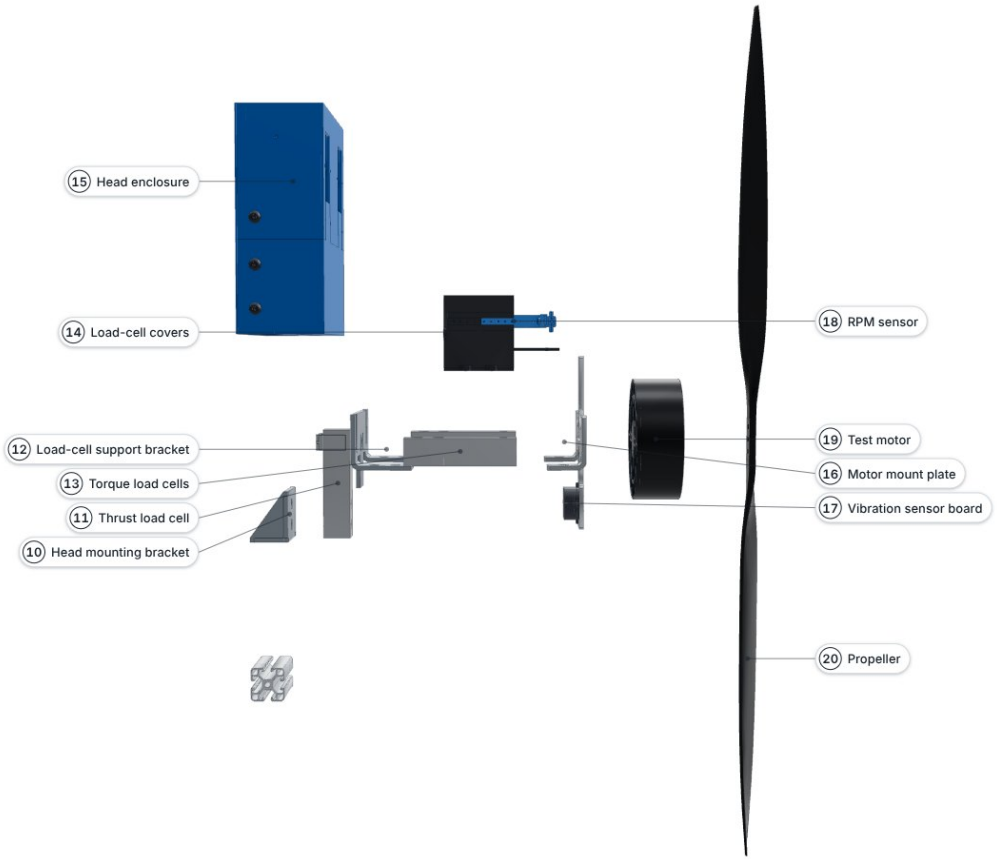


Figure 6.11 Exploded side view of the sensing head, motor and propeller.

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7. DAQ and electronics assembly

The ROTRIX DAQ unit mounts on the rear of the middle pillar bridge, between the pillars and well behind the propeller. The current sensor module sits on the outer face of the right pillar, below the handle, so the battery lead stays clear of the signal cables.

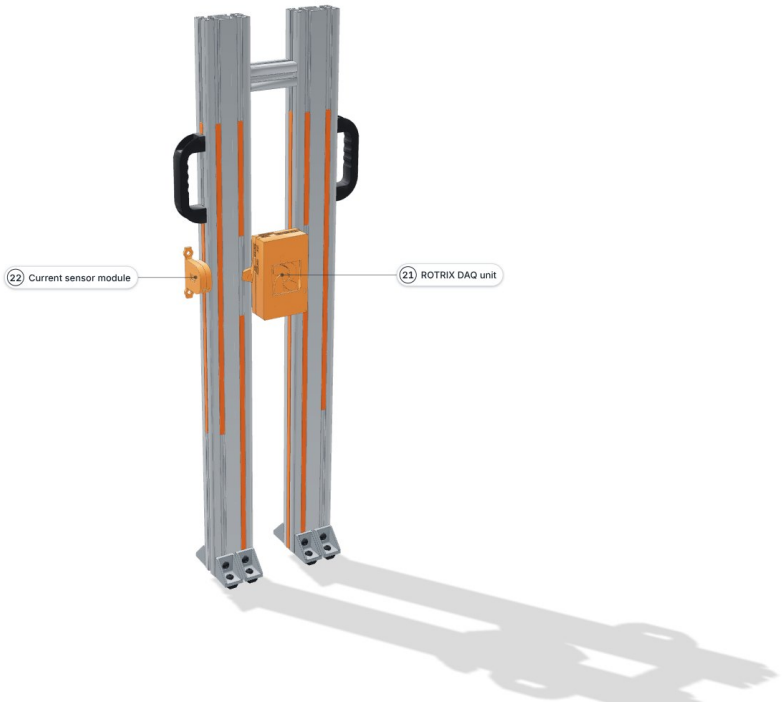


Figure 7.1 ROTRIX DAQ unit on the middle pillar bridge, and the current sensor module on the right pillar.

Mounting and connecting

1. Slide two T-nuts into the rear slot of the middle bridge and screw the DAQ unit on, connectors facing up.
2. Fit the current sensor board in its case on the right pillar, close the cover and screw it shut.
3. Run the battery-to-ESC lead through the current sensor, with the arrow pointing from the battery to the ESC.
4. Connect the thrust cell, the two torque cells, the RPM sensor, the vibration board and the current sensor to the matching DAQ inputs.
5. Connect the DAQ unit to the PC with the USB cable and power it up.

What the DAQ unit measures

- Thrust and torque from the load cells, through precision bridge amplifiers.
- Motor speed from the optical RPM sensor.
- Battery voltage and current in the power line to the ESC.
- Vibration and temperature from the sensor boards.
- All channels are time-stamped together and streamed to ROTRIX for live display, logging and reports.

CAUTION Connect and disconnect sensor cables only with the DAQ unit powered off and the battery disconnected.

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8. Wire routing to the sensors

Good cable routing matters as much as good mechanics. A cable that is pulled tight across the sensing head carries part of the load, and the load cells then measure the cable instead of the motor. Follow the routes and rules below.



Figure 8.1 Recommended cable routes. Signal cables and power leads run on separate sides of the frame.

	Cable
	Thrust load cell
	Torque load cells
	RPM sensor
	Vibration sensor
	Current sensor
	Battery power (through the current sensor)
	USB to PC

Routing table

Cable	From	Route	To
Thrust load cell	Upright cell under the head	Service loop at the cell, down between the pillars in front of the middle bridge, over the bridge	DAQ thrust input
Torque load cells	Both sides of the head	Service loop at each cell, down the front face of each pillar	DAQ torque inputs

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Cable	From	Route	To
RPM sensor	Arm on the left load-cell cover	Back along the arm, down the front face of the left pillar	DAQ RPM input
Vibration sensor	Board behind the motor plate	Service loop, down the front of the head enclosure and between the pillars	DAQ vibration input
Current sensor signal	Module on the right pillar	Short lead round the right pillar to the DAQ unit	DAQ current input
Power leads	Battery at the front right	Up the right pillar, through the current sensor, past the handle to the ESC beside the head	ESC, then motor
USB	DAQ unit	Down the centre and along the floor behind the stand to the operator station	PC

Routing rules

1. **Leave a service loop.** Where a cable leaves the sensing head, form a loose loop about 50 mm across, so the cable carries no load when the head deflects.
2. **Tie to the fixed frame only.** Fix cables to the frame with ties about every 150 mm. Never tie a cable to the motor plate, the load cells or the free end of the sensing head.
3. **Keep signal and power apart.** Run sensor cables and power leads on opposite sides of the frame. Where they must cross, cross at a right angle.
4. **Stay out of the propeller disc.** Route every cable behind the motor plate and away from the plane of the propeller.
5. **Relieve strain at the DAQ.** Leave a little slack at each connector and label both ends of every cable.
6. **Screened load-cell cables.** Connect the screen at the DAQ end only.
7. **Remote operation.** If the operator sits away from the stand, extend only the USB data link and the emergency-stop loop. Never extend the analogue sensor cables.
8. **Current sensor direction.** Fit the current sensor with its arrow pointing from the battery towards the ESC.
9. **Use the slots.** On the fixed frame, lay cables inside the T-slots and press the orange slot covers over them. This keeps cables tidy, protected and clear of the propeller.

CAUTION After any change to the wiring, zero the load cells again and check that the zero reading stays steady for one minute before testing.

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9. Commissioning and calibration

Before first power-up

- All bolts tightened to the recommended torque, and the base anchored.
- Motor fitted and centred; propeller not yet fitted.
- All cables routed with service loops, and the emergency stop tested.

Connect and check the sensors

1. Connect the DAQ unit to the PC over USB and open ROTRIX.
2. Confirm that every sensor is detected and shows a sensible live value.
3. Allow 10 minutes for the load cells to warm up, then zero (tare) thrust and torque with the motor fitted and stationary.

Check thrust and torque with known loads

1. Thrust: tie a cord to the motor hub, run it over a low-friction pulley in line with the thrust axis, and hang known weights (2, 5 and 10 kg). Compare each reading with the weight.
2. Torque: fit a light lever arm of known length to the motor plate and hang a known weight from its end. The expected torque is weight × 9.81 × arm length (N·m, with the weight in kg and the length in m).
3. If any reading is outside the stated accuracy, run the calibration routine in ROTRIX and repeat the check.

Check RPM, voltage and current

- Compare the RPM reading with a hand tachometer at two or three throttle settings.
- Compare voltage and current with a multimeter and a clamp meter.

First run

1. Fit and balance the propeller, and clear the exclusion zone.
2. Connect the battery last. Arm the ESC from the operator station.
3. Run at about 10% throttle for 30 seconds, watching vibration, temperature and noise.
4. Increase in steps of 10% only while every reading stays stable.

10. Operating the stand

- Do not exceed 20 kgf of thrust. Set a thrust limit alarm in ROTRIX below the rating.
- Run a short warm-up before each test series and zero the load cells with the motor stopped.
- Use ROTRIX to run automated throttle steps and ramps, and ROTRIDASH to compare runs and produce PDF performance reports.
- Allow the motor and ESC to cool between long runs, and watch the temperature channels.
- After testing, disarm the ESC, disconnect the battery, then stop logging and save the data.

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11. Maintenance

When	Check
Before every session	Bolts on the sensing head and motor; propeller condition and balance; cable routing and service loops; clamps anchored; emergency stop works
Weekly in regular use	All frame bolts at the recommended torque; sensors clean; no cracks in brackets or printed parts
Every month	Zero and span check of the thrust and torque readings with known weights; RPM check against a hand tachometer
Every 6 months	Full calibration in ROTRIX; inspect cables and connectors; replace worn fasteners
After a crash or overload	Stop testing. Inspect the load cells and brackets, then run a full calibration before further use

NOTE Clean the sensors and the motor plate with a dry cloth after each session. Do not use solvents on the load cells or the circuit boards.

12. Troubleshooting

Symptom	Likely cause	What to do
Thrust reading drifts with the motor stopped	Load-cell cable under tension, or the load cells are still warming up	Re-route the cable with a loose service loop, wait 10 minutes after power-up, then zero again
Readings are noisy or jumpy	Loose bolts, an unbalanced propeller, or signal cables running beside power leads	Retighten the frame, balance the propeller, separate signal cables from the power leads
Torque reads a value with no load	Sensor block not square to the thrust axis, or the motor plate is touching the frame	Check alignment and clearances, then zero again
RPM shows zero or jumps	Sensor not aimed at the blade path, or too far away	Re-aim the sensor and move it closer, keeping it clear of the blade tips
DAQ unit not found by ROTRIX	USB cable, port or driver	Try another cable and port, check the driver, and restart ROTRIX
The stand creeps or rocks during a run	Clamps not anchored, or the floor is uneven	Stop, level the base and anchor every clamp

13. Support and contact

Contact	Details
Sales and quotations	business@reude.tech · cmo_ira@reude.tech
Technical support	support.rotrix@reude.tech
Phone	+91 6384667822
Web	rotrix.reude.tech · www.reude.in
Address	REUDE Technologies Private Limited, No.1, Nehru Street, Vasantham Nagar, Avadi, Chennai 600071, Tamil Nadu, India

When contacting support, quote the document ID on the cover, the stand serial number, and the ROTRIX software version.

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